

Unlearning or Not: A Strategic Data Forgetting Scheme for Federated Unlearning With Bounded Rationality

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Abstract—Federated unlearning (FUL) is a promising approach that enables users to remove the influence of their data from federated learning (FL) models, thereby safeguarding their right to be forgotten and protecting data privacy. However, current FUL studies focus predominantly on designing effective unlearning techniques, with limited attention given to incentivizing client participation while curbing selfish behaviors. Additionally, existing FUL studies often assume that the FL server is a fully rational participant, overlooking the bounded rationality that the FL server might demonstrate when faced with risks and uncertainties. To tackle the aforementioned challenges, we propose a strategic data forgetting scheme with bounded rationality for FUL. Specifically, we first present a practical FUL framework where a boundedly rational FL server permits clients to perform partial data revocation. Then, to incentivize clients to retain more data during the unlearning process, we develop a truthful incentive mechanism based on a prospect theory (PT)-driven data forgetting game, which analyzes the FL server's optimal incentive strategy and the clients' corresponding data revocation decisions. By employing PT, the boundedly rational decision-making process of the FL server is effectively modeled, capturing its behavioral responses under conditions of uncertainty and risk. Furthermore, a penalty mechanism is proposed to mitigate clients' selfish behavior, complemented by a zero-payment mechanism designed to ensure equitable distribution of rewards among clients. Finally, simulation results confirm the efficacy of the proposed scheme in motivating clients to provide high-quality models, suppressing selfishness, and enhancing the accuracy of unlearned models.

Index Terms—Federated unlearning, prospect theory-based data forgetting game, truthful incentive mechanism, penalty mechanism, zero-payment mechanism.

I. INTRODUCTION

FEDERATED learning (FL) as a promising distributed machine learning (ML) paradigm, enables multiple users to collaboratively train a shared model under the coordination

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of a central server without transmitting their local data to the server [2], [3]. By ensuring that only model parameters or gradients are exchanged during training, FL fundamentally reduces the risk of data privacy breaches, which are common in traditional ML approaches due to frequent data exchanges and uploads [4]. Nevertheless, FL may fall short of addressing users' growing privacy demands, particularly the "right to be forgotten" (RTBF) [5]. RTBF allows users to request the removal of their private data and its influence on the trained model when they no longer wish to participate in the platform. Users may choose to leave for various reasons. For instance, users might perceive that the benefits they gain from the platform are insufficient to outweigh their privacy costs. RTBF has been explicitly incorporated into legal frameworks such as the European Union's General Data Protection Regulation (GDPR) [6] and the California Consumer Privacy Act (CCPA) [7], which mandate platforms to enable users to erase their data and its impact from models upon request, thereby safeguarding user privacy.

To uphold the RTBF of users, a straightforward solution involves retraining the model from scratch using the remaining users' data [8]. However, this approach is both computationally expensive and time-consuming, posing significant challenges in practical applications. To address these issues, federated unlearning (FUL) has emerged as an innovative approach to ensure the RTBF of users in FL [9]. Instead of relying on costly retraining procedures, FUL focuses on selectively removing the influence of specific user data from a trained FL model while maintaining the performance of the global model [10]. This process generally involves staying users conducting additional computations based on the trained FL model along with the remaining data to derive an unlearned model.

A critical challenge in FUL lies in incentivizing the participation of remaining clients to mitigate the influence of data from the target clients who will eventually withdraw. Assuming that all remaining clients will altruistically participate in the retraining process is unrealistic. In practice, this process imposes additional costs on these staying clients, including computational and privacy risk costs. Such burdens significantly diminish their willingness to contribute to FUL. Therefore, the development of an effective incentive scheme is essential for motivating the participation of remaining clients.

Extensive studies have focused on developing incentive schemes in FUL to encourage client participation during the unlearning process. For example, in [11], a novel incentive

scheme based on a four-stage Stackelberg game is proposed in a FL and unlearning framework, aiming to incentivize the retention of valuable users and bolster system efficiency. In [12], a two-stage incentive scheme is developed within FUL, employing an evolutionary game and a reinforcement learning approach to address data imbalance and enhance model accuracy. In [13], an incentive scheme within the unlearning framework is introduced, in which a Stackelberg game model is formulated to optimize data revocation pricing and enhance the system's social welfare.

Prior works on FUL have made valuable progress in designing effective incentive schemes. However, they commonly omit several practical aspects. Specifically, most existing studies focus solely on scenarios where users entirely revoke their data, neglecting situations where users may prefer to partially revoke specific portions of their data. This limitation reduces the practical applicability of such schemes. Furthermore, these studies typically assume the FL server acts as a fully rational decision-maker¹, overlooking the bounded rationality² exhibited by the FL server when facing risks and uncertainties. For example, in intelligent healthcare scenarios, the FL server is constrained by limited computational resources, making it difficult to accurately evaluate the true contribution of local models submitted by different institutions to the global model. Under such conditions, the FL server may assign disproportionately high subjective weights to low-probability risk events (e.g., the presence of poisoned updates), while assigning subjective weights lower than their objective probabilities to high-probability routine gains. Finally, the majority of these studies fail to consider penalty mechanisms for clients exhibiting selfish behavior, specifically those who provide meaningless models. The lack of penalty mechanisms will diminish the willingness of honest clients to participate and undermine the effectiveness of FUL. Therefore, considering the bounded rationality of the FL server, designing an incentive scheme in FUL to guide remaining clients in selecting optimal data revocation strategies while suppressing the selfish behavior of clients remains an open challenge.

To address the aforementioned issues, we propose a strategic data forgetting scheme with bounded rationality for FUL that aims to motivate user participation while curbing their selfishness. Specifically, we first design a practical FUL framework where the boundedly rational FL server allows clients to perform partial data revocation. Then, we propose a truthful incentive mechanism in which the interactions between the FL server and clients are modeled as a prospect theory (PT)-based data forgetting game, with PT used to capture the decision-making process of the boundedly rational FL server. In this game, the FL server acts as the leader, formulating a set of optimal recruitment strategies that include the remaining data size and

the corresponding rewards. Each client, as a follower, decides on the optimal data revocation strategy based on the reward strategy established by the FL server. Subsequently, the FL server uses a remaining data size evaluation algorithm to assess the model updates uploaded by each remaining client, enabling the estimation of the remaining data size for each client and allocating the corresponding rewards accordingly. Furthermore, to address the issue of selfish clients who provide meaningless models, we devise a penalty mechanism to suppress selfish behaviors. Finally, to ensure fairness in reward allocation, we design a zero-payment mechanism to penalize clients who ultimately opt out of the unlearning process. The contributions of this work are three-fold:

- *A practical FUL framework with partial data revocation support:* We develop a practical FUL framework that accounts for the bounded rationality of the FL server, enabling clients to revoke their data partially. In this framework, a truthful incentive mechanism is designed to encourage clients to retain more data during the unlearning process while curbing their selfish behavior.
- *PT-based data forgetting game modeling:* We introduce a PT-based data forgetting game to analyze the optimal incentive for the FL server and the corresponding data revocation strategies for clients. In this game, PT is leveraged to model the decision-making process of the boundedly rational FL server, reflecting its responses under uncertainty and risk. We design a penalty mechanism to suppress clients' selfish behavior and a zero-payment mechanism to ensure fairness in reward distribution.
- *Performance evaluation:* Through extensive simulation experiments, we evaluate the efficiency of the proposed scheme. Numerical results indicate that our scheme can effectively suppress selfish behaviors of clients, incentivize them to provide high-quality models, and enhance the utilities of both the FL server and the clients.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III presents the preliminaries. The system model is described in Section IV. The truthful incentive mechanism is elaborated in Section V. The equilibrium of the PT-based data forgetting game is presented in Section VI. The experimental results and performance analysis are given in Section VII. Section VIII concludes the paper.

II. RELATED WORK

A. Federated Unlearning

Recently, FUL has garnered considerable attention in academia. Liu et al. [8] proposed a privacy-preserving FUL scheme that integrates the secure aggregation protocol to safeguard users' data privacy throughout the FUL process while efficiently dealing with dynamic users' strategies. Gao et al. [9] introduced a FUL framework that combines multiple unlearning and verifiable approaches, which enables a quantitative assessment of unlearning effects. Zhang et al. [10] presented a FUL algorithm based on differential privacy, which removes a client's influence from the unlearned global model by subtracting a weighted sum of gradient residuals and crops noise to ensure

¹The principle of rationality serves as a basic assumption of economics and game theory. It posits that participants, free from the influence of subjective biases, consistently select strategies that maximize their utility to achieve optimal economic outcomes.

²In game theory, bounded rationality refers to the existence of decision-making psychological preferences of participants when confronted with subjective interference. Irrationality refers to participants' decisions that are random and cannot be predicted.

statistical indistinguishability between the unlearned model and the retrained model. Wang et al. [14] investigated a server initiating FUL scheme to eliminate the impact of low-quality data on the global model, where a category-based approach was devised to identify clients contributing low-quality data, thereby enhancing the performance of the unlearned model. Xia et al. [15] developed a FUL method for digital twin environments that achieves accurate data forgetting while improving model accuracy through memory evaluation and multi-loss training methods.

However, current research in FUL predominantly operates under the assumption of client willingness to actively engage in the unlearning process, ignoring the reality that clients may refrain from participating in FUL in the absence of sufficient incentives.

B. Incentive Scheme for Federated Unlearning

Extensive research has concentrated on the incentive schemes for FUL. Shao et al. [5] proposed an incentive scheme based on the Stackelberg game in FUL, aiming to strategically guide clients towards strategies that preserve system stability. Ding et al. [11] developed an incentive scheme to encourage the retention of valuable users within the system for FL and unlearning. Moreover, the authors applied a four-stage Stackelberg game to derive optimal strategies for participants, aiming to enhance their utility. Lin et al. [12] presented a two-stage incentive scheme for FUL, which motivates high-quality clients to engage in unlearning by applying an evolutionary game and a reinforcement learning approach, thus solving the problem of imbalanced data distribution and improving model accuracy. Cui et al. [13] introduced an incentive scheme in an unlearning framework that aims to offset the cost of the server in the unlearning process. Furthermore, the authors formulated a Stackelberg game to optimize data revocation pricing to maximize the social welfare of the system. Despite the extensive research focused on incentive schemes in FUL, the majority of studies assume that the FL server is a fully rational decision-maker, ignoring the situation where the FL server may demonstrate bounded rationality in the face of risky and uncertain environments.

Unlike them, by employing PT to model the decision-making process of the FL server with bounded rationality, we propose an incentive scheme grounded in the PT-based data forgetting game to motivate clients to contribute high-quality models in unlearning.

III. PRELIMINARIES

A. Federated Learning

In FL, a set of clients, denoted as $\mathcal{I} = \{1, \dots, i, \dots, I\}$, engage in collaborative training to develop a globally shared model by utilizing their local data. In this context, each client i possesses a local training dataset $\mathbb{D}_i$, which contains a number of $D_i = |\mathbb{D}_i|$ samples (i.e., training data). Here, $\mathbb{D}_i$ contains D_i input-output pairs (ϱ_j, γ_j) , $1 \leq j \leq D_i$, where ϱ_j represents the input data, and γ_j denotes the output value of ϱ_j . The primary goal of FL is to find the optimal model parameters, denoted as

ϖ , which minimizes the global loss function $\mathcal{L}(\varpi)$ across the data set, i.e.,

$$\varpi^* = \arg \min_{\varpi} \mathcal{L}(\varpi) = \arg \min_{\varpi} \frac{1}{I} \sum_{i=1}^I \mathcal{L}_i(\varpi), \quad (1)$$

where $\mathcal{L}_i(\varpi)$ denotes the local loss function of client i associated with local training dataset $\mathbb{D}_i$, defined as

$$\mathcal{L}_i(\varpi) = \frac{1}{D_i} \sum_{j=1}^{D_i} \ell_j(\varpi), \quad (2)$$

where $\ell_j(\varpi)$ is the loss function associated with data sample (ϱ_j, γ_j) , which quantifies the error of model ϖ with respect to data sample (ϱ_j, γ_j) .

B. Federated Unlearning

FUL is an emerging privacy-preserving machine learning technique that achieves the effect of having the global model forget data from leaving clients by leveraging the storage of model updates [16]. Unlike the FL process, which maps clients' data into the model space, FUL maps clients' data, a learned model, and the datasets to be forgotten into the unlearning model space. The aim of FUL is to ensure that the unlearning model exhibits a distribution equivalent to that of a model retrained from the outset with the remaining data.

A typical approach in the context of FUL involves allowing the staying clients to continue training from the learned model using their remaining data [17]. The training process is iteratively conducted until the unlearning model converges to a new model parameter that minimizes the global loss function for the staying clients [18], i.e.,

$$\tilde{\varpi}^* = \arg \min_{\varpi} \frac{1}{I'} \sum_{i \in \mathcal{I}'} \mathcal{L}_i(\varpi), \quad (3)$$

where $\mathcal{I}'$ denotes the set of clients who remain in the system through FUL. I' is the number of clients who remain in the system through FUL. From (3), we find that the unlearned model is significantly influenced by the data revocation strategies for clients.

IV. SYSTEM MODEL

In this section, we introduce the system model, which consists of the network model and the remaining data size assessment model. For improved readability, the principal symbols employed throughout this paper are summarized in Table I.

A. Network Model

As illustrated in Fig. 1, we consider a typical FUL system wherein a set of clients have participated in FL by utilizing their local datasets under the orchestration of the FL server, denoted as S . These heterogeneous clients are now confronted with a FUL decision, i.e., simultaneously determining the amount of data to be revoked from the learned model, thereby ensuring the efficient completion of the unlearning process. Let d_i represent the data size remaining in the system for each client i after deciding on data revocation. In this setup, the amount of data revoked by client i through FUL is $D_i - d_i$. Upon receiving data

TABLE I
SUMMARY OF NOTATIONS

Notations	Descriptions
$\mathcal{I}$	The set of clients participating in the previous FL.
$\mathcal{I}'$	The set of clients participating in FUL.
T	The number of unlearning rounds.
d_i	The retained data size of client i .
$\hat{d}_i$	The perceived retained data size of client i by the server.
$\mathbf{d}$	The vector of the remaining data size from staying clients.
$\mathbf{r}$	The reward vector of the FL server.
λ	The evaluation error of the FL server.
θ_i	The weighting parameter for the FL server's satisfaction.
τ_i	The data uniqueness level of client i .
$\varphi_{i,1}$	The marginal privacy cost of client i .
$\varphi_{i,2}$	The marginal training cost of client i .
α	The objective weight of the FL server.

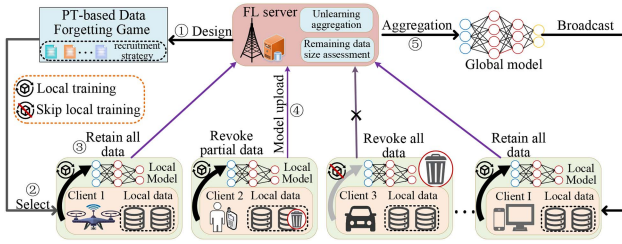


Fig. 1. Illustration of the proposed truthful incentive framework for federated unlearning.

revocation requests from the clients, the FL server orchestrates the staying clients to continue training from the learned FL model, utilizing the remaining data. Moreover, the FL server maintains the local model updates submitted by each client in each global iteration of the previous FL. These updates serve as valuable priori knowledge, which can be leveraged to estimate the data size revoked by each client through FUL.

However, clients incur various costs, including energy costs, during the unlearning process, which may lead them to revoke more data through FUL or opt out of the unlearning process. To motivate clients to engage in FUL, we propose a reward-based solution wherein the FL server provides compensation to clients for the costs incurred during the unlearning. A dual-objective dilemma arises in this setting: (i) the FL server seeks to stimulate clients to retain more data through FUL with minimal payments, thereby enhancing the performance of the unlearned model. (ii) Clients aim to receive higher rewards from the FL server for their participation in the unlearning process. Moreover, decisions made by the FL server can deviate from the predictions of expected utility theory (EUT) due to the interference of subjective factors, including psychological perceptions.

In this paper, we propose a truthful incentive scheme based on PT to address the above challenges. The specific workflow of the proposed scheme is as follows:

- Initially, upon receiving the unlearning request, the FL server employs a PT-driven data forgetting game to design a set of recruitment messages, which enumerate the reward associated with the remaining data size for unlearning, for clients previously involved in FL. Subsequently, the FL server broadcasts these recruitment messages to clients.

- Based on the reward strategy of the FL server, each client determines the amount of data to be revoked through FUL to maximize its utility.
- When certain clients ultimately decide to leave the system (i.e., revoke all data), the FL server removes the historical model updates associated with these leaving clients. Then, the FL server retrieves the most recent global model in the previous FL in which these leaving clients did not participate and subsequently broadcasts it to staying clients.
- Next, according to the selected strategy, each staying client utilizes its remaining data for unlearning. Afterwards, each staying client transmits the trained local model to the FL server.
- Finally, the FL server aggregates the results provided by staying clients and disseminates the aggregated results to them for subsequent unlearning. Besides, the FL server compensates each staying client based on the data size revoked. The training process is iteratively conducted until the unlearning model converges to a new model parameter that minimizes the global loss function for the staying clients.

B. Remaining Data Size Assessment Model

In this subsection, we introduce a remaining data size assessment model to estimate the data size remaining in the system for each staying client. Specifically, at the beginning of the FUL process, the FL server instructs each staying client i to first execute a single unlearning round and then upload the updated model parameters $\tilde{\omega}_i^0$. Subsequently, the FL server retrieves each client i 's model updates ω_i^k in the k -th round of prior FL (assuming the client that is ultimately to be retired joins prior FL in round k). In this context, based on the influence function-based algorithm [19], the FL server can estimate the remaining data size for client i by comparing the magnitude of model updates before and after forgetting. In statistics, the influence function shows that, under a first-order approximation, the change in model parameters resulting from an infinitesimal perturbation to the weight of a training sample (or the removal of a single sample) can be characterized by the product of the inverse Hessian matrix and the sample's gradient. This establishes a principled connection between “adding or removing samples” and the induced “changes in model parameters or gradients”³ [20]. Furthermore, the independently and identically distributed (IID) assumption is widely adopted to ensure that the influence function can mathematically map parameter perturbations to the corresponding volume of retained data [19], [20]. Thus, consistent with these studies, we assume that the clients' data are IID. Under this assumption, the first-order

³The core contribution of this work is to design a trustworthy incentive scheme that encourages users to retain as much data as possible in the FUL. Therefore, we focus solely on the size of the retained data. To isolate the effect of data removal from other training dynamics, we assume fixed gradient-related factors, including learning rate, batch order, optimizer state, and random seeds. This establishes a controlled environment in which observed parameter perturbations are exclusively attributable to data removal rather than stochastic variations. This approach aligns with the methodology in existing strategic FUL and game-theoretic incentive studies [12], [18].

approximation implies that removing $\mathcal{M}$ samples yields a gradient shift that is statistically proportional to $\mathcal{M}$, given that the learning rate is constant. Since a stochastic gradient is the empirical average of per-sample gradients, the magnitude of the resulting parameter deviation grows approximately linearly with the number of removed samples [21]. This observation provides theoretical support for employing the influence function to estimate the size of the remaining data. Let σ be the minimizer or an approximate convergence point of the empirical loss $\mathcal{L}(\varpi)$. Then, after removing a single sample χ , the first-order approximation of the resulting parameter change is given by

$$\sigma_{-\chi} = \sigma + \frac{1}{D_i} \mathcal{H}_\sigma^{-1} \nabla_\sigma \mathcal{L}_i(\chi; \sigma), \quad (4)$$

where $\mathcal{H}$ denotes the empirical Hessian matrix. If $\mathcal{M}$ samples with similar gradient characteristics are removed, the overall parameter perturbation can be approximated as

$$\Delta\sigma \propto \frac{\mathcal{M}}{D_i}. \quad (5)$$

By mapping this parameter change into the gradient space, we observe that the magnitude of the update difference is approximately proportional to the amount of data removed. Therefore, we use the influence function, which captures both the magnitude of the update differences, to assess the client's remaining data size. Under non-IID conditions, the FL server can leverage a representation-layer difference-based method [22] to evaluate the size of clients' remaining data. Specifically, the FL server leverages a multi-task information bottleneck architecture to assess the remaining data size by extracting differences within the model's representation layer⁴.

For simplicity, the remaining data size is quantized into $H + 1$ levels, where each staying client i determines the remaining data size $d_i \in \{0, 1, \dots, H\}$. In addition, we introduce an opt-out state $d_i = -1$ to denote that client i chooses not to participate in FUL (i.e., retains no data). Hence, $d_i \in \{-1, 0, \dots, H\}$. Herein, $d_i = 0$ signifies that client i contributes meaningless local model. $d_i = H$ refers to the maximum remaining data size. Let $\Pr(\hat{d}|d)$ denote the probability that the FL server perceives the actual remaining data size level d as $\hat{d}$. We assume that under good wireless channel conditions, the FL server is capable of receiving the local models transmitted by clients. This implies that the FL server possesses the capability to accurately ascertain the client's participation status in FUL. Consequently, we have $\Pr(-1|-1) = 1$. In addition, the FL server may encounter inaccuracies when evaluating the remaining data size. Thus, the probability that a local model with remaining data size level d is perceived as $d \pm 1$ (if applicable) is $\lambda/2$. Here, $\lambda \in (0, 1)$ represents the evaluation error of the FL server, which is influenced by the efficacy of the utilized evaluation algorithm, the amount of available computational and storage resources, and

the number of remaining data size levels. When the client contributes meaningless local models (i.e., $d = 0$) or does not intend to revoke data (i.e., $d = H$), the probability that the FL server commits an evaluation error in assessing the remaining data size are represented by $\Pr(1|0) = \lambda$ and $\Pr(H-1|H) = \lambda$, respectively. Consequently, based on [23], the probability mass function for the remaining data size assessment model is given by

$$\Pr(\hat{d}|d) = \begin{cases} 1 - \lambda, & \text{if } \hat{d} = d \geq 0, \\ \lambda, & \text{if } (d, \hat{d}) = (0, 1) \text{ or } (H, H-1), \\ \lambda/2, & \text{if } \hat{d} = d \pm 1, 0 < d < H, \\ 1, & \text{if } (d, \hat{d}) = (-1, -1), \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

For the sake of analysis, we presuppose that H and λ are known and fixed. Note that the proposed framework is not restricted to the remaining data assessment model in (6), as the analytical results in Section VI are independent of (6).

V. TRUTHFUL INCENTIVE MECHANISM

In this section, we propose a truthful incentive mechanism to encourage staying clients to contribute high-quality models (i.e., to revoke a small amount of data through FUL) while curbing the self-interested behaviors of clients. Specifically, we first formulate the interactions between the FL server and staying clients as a PT-based data forgetting game. Then, we model the utility functions of both the FL server and clients.

A. PT-Based Data Forgetting Game Formulation

During the execution of the FUL task, both the FL server and staying clients aim to maximize their respective benefits. This leads to a conflict of interest between the FL server and staying clients. Specifically, the FL server seeks to motivate clients to retain more data with minimal payments, while each staying client aims for the highest possible payment. To resolve the conflicting goals, we formulate the interactions between the FL server and staying clients as a PT-based data forgetting game. Besides, to ensure fairness, clients are eligible to earn rewards solely for participation in FUL. Furthermore, to discourage clients' selfish behavior, we design a penalty mechanism that entails imposing fines on clients who contribute meaningless models.

We define the PT-based data forgetting game in the following Definition 1.

Definition 1 (PT-Based Data Forgetting Game): In the FUL process, the interactions between the FL server and staying clients are modeled as a PT-based data forgetting game, i.e.,

$$\mathcal{G} = \{\{\mathcal{I}', S\}, \{\mathbf{d}, \mathbf{r}\}, \{\mathbb{U}_S^{PT}(d_i, \mathbf{r}), \mathbb{U}_i^{PT}\}_{i \in \mathcal{I}'}\}. \quad (7)$$

The PT-based data forgetting game $\mathcal{G}$ comprises the following components:

- **Players:** The players in game $\mathcal{G}$ are (i) staying clients in the set $\mathcal{I}'$ and (ii) the FL server S .

⁴The remaining data assessment model and the PT-driven incentive scheme are decoupled. Specifically, the recruitment strategy (i.e., game equilibrium) is derived in advance, before the unlearning process. After unlearning, the remaining data assessment model is used only to estimate the retained data size for reward allocation. Therefore, non-IID data does not change the equilibrium strategy itself.

- *Strategies*: The strategy of the FL server is to subjectively design a set of recruitment strategies, which enumerate the reward vector $\mathbf{r}$ associated with remaining data size $\mathbf{d}$ to maximize its subjective expected utility. The strategy of each client is to choose a remaining data size level to maximize its subjective expected utility.
- *Utilities*: The subjective expected utilities of each client i and the FL server in game $\mathcal{G}$ are defined as $\mathbb{U}_i^{PT}$ and $\mathbb{U}_S^{PT}(\mathbf{d}, \mathbf{r}) = \sum_{i \in \mathcal{I}'} \mathbb{U}_S^{PT}(d_i, \mathbf{r})$, respectively. Here, $\mathbb{U}_S^{PT}(d_i, \mathbf{r})$ is the subjective expected utility of the FL server for client i .

B. Utility Function

The utility of the FL server: In FUL, the FL server uses the unlearned global model to provide artificial intelligence (AI) services, meaning that models with better performance bring greater benefits to the FL server [24]. According to [25], the model's performance is positively correlated with the data size utilized by clients for model training. Consequently, the satisfaction of the FL server depends on the size of the remaining data on the clients. Moreover, the FL server needs to remunerate each client i according to the evaluated remaining data size $\hat{d}_i$. Hence, the utility of the FL server is the difference between its satisfaction and the reward paid to staying clients, which is defined as

$$\begin{aligned} U_S(\mathbf{d}, \mathbf{r}) &= \sum_{i=1}^{I'} u_S(d_i, \mathbf{r}) \\ &= \sum_{i=1}^{I'} \theta_i G(d_i) - r(\hat{d}_i), \end{aligned} \quad (8)$$

where $\mathbf{d} = [d_1, \dots, d_i, \dots, d_{I'}]$ is the vector of remaining data size from all clients. $u_S(d_i, \mathbf{r})$ represents the utility of the FL server for client i . $G(d_i)$ denotes the satisfaction of the FL server with the actual remaining data size d_i provided by client i . θ_i is the weighting parameter for the FL server's satisfaction. $\mathbf{r} = [r(1), r(2), \dots, r(H)]$ denotes the reward vector of the FL server, which encompasses H non-zero feasible reward levels. $r(\hat{d}_i)$ is the reward of the FL server for the evaluated remaining data size $\hat{d}_i$ provided by client i .

The utility of the client: The utility of client i consists of two parts: the reward paid by the FL server and the cost of performing the unlearning task. The cost of performing unlearning tasks includes both privacy costs and training costs. Thus, the utility of client i is given by

$$u_i(d_i, \mathbf{r}) = r(\hat{d}_i) - C_{i,1}(d_i) - C_{i,2}(d_i), \quad (9)$$

where $C_{i,1}(d_i)$ is the privacy costs, and $C_{i,2}(d_i)$ is the training costs. Similar to [26], we define a client's privacy cost as the economic loss caused by privacy breaches. Intuitively, the privacy cost increases with the amount of remaining data, since a larger data volume provides more evidence for attackers, thereby increasing the likelihood of successful inference. On the other hand, the privacy cost is positively correlated with the level of data uniqueness. The uniqueness level of the data is quantified by the computed gradient, as the gradient characterizes the deviation of client i 's data from the mean of the distribution

pertaining to other clients. For a fixed data size, a higher level of data uniqueness (i.e., greater deviation from other clients) implies a higher probability that attackers can infer private information from the client, thus leading to an increased privacy cost. Therefore, the privacy costs of client i is defined as

$$C_{i,1}(d_i) = \varphi_{i,1} \tau_i d_i, \quad (10)$$

where τ_i is the data uniqueness level of client i . $\varphi_{i,1}$ is the marginal privacy cost of client i .

Based on [27], the training costs across T unlearning rounds are positively related to its remaining data size, which can be expressed as

$$C_{i,2}(d_i) = \varphi_{i,2} T d_i, \quad (11)$$

where $\varphi_{i,2}$ is the marginal training cost of client i . Substituting (10) and (11) into (9), the utility of the client can be rewritten as

$$\begin{aligned} u_i(d_i, \mathbf{r}) &= r(\hat{d}_i) - C_i(d_i) \\ &= r(\hat{d}_i) - \varphi_{i,1} \tau_i d_i - \varphi_{i,2} T d_i, \end{aligned} \quad (12)$$

where $C_i(d_i)$ represents the client i 's total cost with remaining data size d_i . For convenience, we denote $r(\hat{d}_i)$ as $r_{\hat{d}_i}$. With a zero-payment mechanism, the FL server only rewards clients who participate in FUL, thereby ensuring fairness in the distribution of rewards. As such, we have $r_{-1} = 0$. With penalty punishment, the FL server imposes penalties on clients that utilize false data to participate in training to contribute meaningless models (i.e., $d = 0$), so we have $r_0 = -R$. Herein, R is the penalty imposed by the FL server on clients for providing meaningless models. Since $u_i(-1, \mathbf{r}) = 0 > -R - C_i(0) = u_i(0, \mathbf{r})$, it is reasonable to infer that clients would be more motivated to withdraw from FUL rather than participate using false data. Therefore, the proposed scheme ensures fairness among clients while effectively mitigating their selfish behavior⁵.

Expected utility of participants: Considering that the FL server may sometimes fail to accurately evaluate remaining data size, if $\Pr(\hat{d} = d)$ is small, a client may receive either excessive or insufficient reward, with the expected reward given by $\sum_{h=-1}^H r_h \Pr(h|d)$ on all potential outcomes of the remaining data size estimation. Consequently, the expected utility of client i is calculated by

$$\begin{aligned} \mathbb{U}_i(d_i, \mathbf{r}) &= \mathbb{E}_{\Pr(\hat{d}_i|d_i)} [u_i(d_i, \mathbf{r})] \\ &= \sum_{h=-1}^H r_h \Pr(h|d_i) - C_i(d_i). \end{aligned} \quad (13)$$

⁵In FUL, the presence of colluding malicious clients can indeed threaten the security of the system. Existing research has proposed various schemes for defending against collusion attacks, including vector commitment-based dual-verifiable methods [28] and dimension reduction-assisted robust aggregation algorithms [29]. However, the core contribution of this paper is the design of a trustworthy incentive scheme. Addressing potential collusion attacks of clients falls outside the scope of this paper. Therefore, consistent with the commonly adopted assumptions in existing incentive schemes for FUL [11], [18], we assume that clients are non-colluding.

The expected utility of the FL server from I' clients can be formulated as

$$\begin{aligned} \mathbb{U}_S(\mathbf{d}, \mathbf{r}) &= \sum_{i=1}^{I'} \mathbb{E}_{\Pr(\hat{d}_i|d_i)} [u_S(d_i, \mathbf{r})] \\ &= \sum_{i=1}^{I'} \left[\theta_i G(d_i) - \sum_{h=-1}^H r_h \Pr(h|d_i) \right]. \end{aligned} \quad (14)$$

Subjective expected utility of participants: Considering the impact of subjective elements and psychological perceptions [30], the FL server, as an intelligent decision-making entity, does not always behave rationally. Specifically, in the context of risky and uncertain environments, the FL server may be inclined to make subjective decisions regarding the estimation of the remaining data size, which may diverge from the outcomes predicted by EUT. Consequently, we utilize PT in the data forgetting game to formalize the FL server's subjective decision-making process. In PT, participants tend to underestimate the likelihood of high-probability events and overestimate that of low-probability events, which is referred to as the probability weighting distortion effect [31]. Thus, based on the Prelec weighting function [32], the subjective probability mass function for the remaining data size assessment model is defined as

$$\omega(\Pr(\hat{d}|d)) = \exp\left(-\left(-\ln \Pr(\hat{d}|d)\right)^\alpha\right), \quad (15)$$

where α denotes the objective weight that reflects the level of rationality exhibited by the FL server. By substituting the objective probabilities from the remaining data size assessment model in (14) with the subjective probability in (15), the subjective expected utility of the FL server can be defined as

$$\begin{aligned} \mathbb{U}_S^{PT}(\mathbf{d}, \mathbf{r}) &= \sum_{i=1}^{I'} \mathbb{U}_S^{PT}(d_i, \mathbf{r}) \\ &= \sum_{i=1}^{I'} \left[\theta_i G(d_i) - \sum_{h=-1}^H r_h \omega(\Pr(h|d_i)) \right]. \end{aligned} \quad (16)$$

Similarly, the subjective expected utility of client i can be formulated as

$$\mathbb{U}_i^{PT}(d_i, \mathbf{r}) = \sum_{h=-1}^H r_h \omega(\Pr(h|d_i)) - C_i(d_i). \quad (17)$$

In the PT-based data forgetting game, the goal of the participants is to determine the optimal strategies to maximize their subjective expected utility⁶. Specifically, the FL server, as the leader, first designs a set of recruitment messages, which enumerate the reward associated with the remaining data size, with the aim of maximizing its subjective expected utility. Each client i , $\forall i \in \mathcal{I}'$ as a follower chooses the optimal remaining data size d_i to maximize its subjective expected utility. We then introduce the equilibrium of the PT-based data forgetting game in the following Definition 2.

Definition 2: The equilibrium for the PT-based data forgetting game is denoted as $(\mathbf{d}^*, \mathbf{r}^*)$, which consists of the optimal

response for each player in the game. By definition, we have⁷

$$d_i^*(\mathbf{r}) = \arg \max_{-1 \leq d_i \leq H} \mathbb{U}_i^{PT}(d_i, \mathbf{r}), \forall i \in \mathcal{I}'. \quad (18)$$

$$\mathbf{r}^* = \arg \max_{\mathbf{r} \geq 0} \mathbb{U}_S^{PT}(\mathbf{d}^*(\mathbf{r}), \mathbf{r}). \quad (19)$$

The proposed PT-based data forgetting game aims to find an equilibrium in which both the clients and the FL server can select their optimal strategies to maximize their subjective expected utilities. We first analyze the scenario with a single positive remaining data size, referred to as Scenario I. In this scenario, client i decides the maximum data size to remain in the system with $d_i = 1$, contributes a meaningless local model with $d_i = 0$, and decides to revoke all data through FUL with $d_i = -1$. Subsequently, we further consider the equilibrium of the PT-based data forgetting game with $H = 2$ non-zero remaining data size levels, referred to as Scenario II. In this scenario, client i decides the maximum data size to remain in the system with $d_i = 2$, decides the part of the data size to remain in the system with $d_i = 1$ (herein, we set $d_i = 1$ to represent retaining half of the maximum data), contributes a meaningless local model with $d_i = 0$, and decides to revoke all data through FUL with $d_i = -1$.

VI. EQUILIBRIUM OF PT-BASED DATA FORGETTING GAME

A. Game Analysis in Scenario I ($H = 1$)

To obtain the equilibrium for the proposed PT-based data forgetting game, we first analyze the scenario with a single positive remaining data size ($H = 1$). In addition, with zero-payment, the FL server only rewards clients who participate in FUL. As such, $r_{-1} = 0$. Furthermore, to punish clients who provide meaningless models, the penalty punishment mechanism is devised to discourage the selfish behavior of clients. As such, $r_0^* = -R$, so we have $\mathbf{r}^* = [0, -R, r^*]$. We first establish the quantitative correlation between the optimal reward strategy of the FL server and the error in assessing the remaining data size in Proposition 1.

Proposition 1: In the PT-based data forgetting game with $H = 1$ positive remaining data size levels, client i is stimulated to decide the maximum data size to remain in the system, i.e., $d_i = 1$ if $\lambda < \frac{1}{2}$ and

$$\begin{aligned} r^* &> \max \left\{ \frac{[\omega(\lambda) - \omega(1 - \lambda)] R + C_i(1) - C_i(0)}{\omega(1 - \lambda) - \omega(\lambda)}, \right. \\ &\quad \left. \frac{\omega(\lambda) R + C_i(1)}{\omega(1 - \lambda)} \right\}. \end{aligned} \quad (20)$$

⁶Under EUT, the FL server directly utilizes objective probabilities to maximize its expected utility and derive the optimal reward allocation. In contrast, our work models the FL server's bounded rationality through the probability weighting effect in PT. Therefore, the FL server optimizes its subjective expected utility based on subjective probabilities rather than objective probabilities.

⁷The FL server is responsible for estimating the remaining data size of clients and determining a set of optimal recruitment strategies. Owing to the inherent uncertainty of the remaining data assessment, its perception of the assessment accuracy may deviate from the objective probability. Therefore, PT is employed to characterize the FL server's bounded rational behavior. In contrast, clients simply select their optimal strategy to maximize their utility based on the set of recruitment strategies provided by the FL server. This requires only a straightforward comparison of all possible strategies, and thus, the clients are modeled as fully rational decision-makers.

Client i is stimulated to decide to revoke all data through FUL, i.e., $d_i = -1$ if

$$r^* \leq \min \left\{ \frac{\omega(1-\lambda)R + C_i(0)}{\omega(\lambda)}, \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right\}. \quad (21)$$

Proof: As $H = 1$, (6) can be reformulated as

$$\Pr(\hat{d}|d) = \begin{cases} 1-\lambda, & \text{if } (d, \hat{d}) = (0, 0) \text{ or } (1, 1), \\ \lambda, & \text{if } (d, \hat{d}) = (0, 1) \text{ or } (1, 0), \\ 1, & \text{if } (d, \hat{d}) = (-1, -1), \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

As $\lambda < \frac{1}{2}$, based on (17) and (22), we have

$$\begin{aligned} \mathbb{U}_i^{PT}(1, r^*) \\ = \omega(1-\lambda)r^* - \omega(\lambda)R - C_i(1) \geq 0 = \mathbb{U}_i^{PT}(-1, r^*), \end{aligned} \quad (23)$$

if

$$r^* \geq \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)}, \forall i \in \mathcal{I}', \quad (24)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(1, r^*) &= \omega(1-\lambda)r^* - \omega(\lambda)R - C_i(1) \\ &\geq -\omega(1-\lambda)R + \omega(\lambda)r^* - C_i(0) = \mathbb{U}_i^{PT}(0, r^*), \end{aligned} \quad (25)$$

if

$$r^* \geq \frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)}, \forall i \in \mathcal{I}'. \quad (26)$$

Thus, combining Formula (24) and Formula (26), we have

$$\mathbb{U}_i^{PT}(1, r^*) \geq \max \{ \mathbb{U}_i^{PT}(-1, r^*), \mathbb{U}_i^{PT}(0, r^*) \}, \quad (27)$$

if

$$\begin{aligned} r^* &> \max \left\{ \frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)}, \right. \\ &\quad \left. \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right\}. \end{aligned} \quad (28)$$

Therefore, if Formula (20) holds, client i is stimulated to decide the maximum data size to remain in the system, i.e., $d_i = 1$.

Similarly, from (17) and (22), we have

$$\begin{aligned} \mathbb{U}_i^{PT}(-1, r^*) \\ = 0 \geq \omega(1-\lambda)r^* - \omega(\lambda)R - C_i(1) = \mathbb{U}_i^{PT}(1, r^*), \end{aligned} \quad (29)$$

if

$$r^* \leq \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)}, \forall i \in \mathcal{I}', \quad (30)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(-1, r^*) &= \\ 0 &\geq -\omega(1-\lambda)R + \omega(\lambda)r^* - C_i(0) = \mathbb{U}_i^{PT}(0, r^*), \end{aligned} \quad (31)$$

$$r^* \leq \frac{\omega(1-\lambda)R + C_i(0)}{\omega(\lambda)}, \forall i \in \mathcal{I}'. \quad (32)$$

Thus, combining Formula (30) and Formula (32), we have

$$\mathbb{U}_i^{PT}(-1, r^*) \geq \max \{ \mathbb{U}_i^{PT}(0, r^*), \mathbb{U}_i^{PT}(1, r^*) \}, \quad (33)$$

if

$$r^* \leq \min \left\{ \frac{\omega(1-\lambda)R + C_i(0)}{\omega(\lambda)}, \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right\}. \quad (34)$$

Thus, if Formula (21) holds, Client i is stimulated to decide to revoke all data through FUL, i.e., $d_i = -1$. Proposition 1 is proved. ■

Remark: Proposition 1 demonstrates that the FL server's optimal reward strategy increases with the error in evaluating the remaining data size to compensate for the underpayment loss of the clients. Additionally, when the reward for high remaining data sizes is too low compared to that in Formula (20) and Formula (21), clients may find it advantageous to engage in self-interested behaviors, leading to an incentive for them to contribute meaningless local models.

Let $\mathbf{1}$ be the vector with all elements of 1. Theorem 1 presents the optimization analysis and proves the equilibrium.

Theorem 1: In the PT-based data forgetting game with $H = 1$ positive remaining data size levels and I' clients, the game equilibrium is given by

$$\mathbf{d}^* = \mathbf{1}_{I'}, \quad (35)$$

$$\mathbf{r}^* = \left[0, -R, \max_{i \in \mathcal{I}'} \max \left(\frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)}, \frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)} \right) \right]. \quad (36)$$

if

$$\begin{aligned} \min \left(\frac{\theta_i G(1)}{\omega(1-\lambda) - \omega(\lambda)} - R, \frac{\theta_i G(1) + \omega(\lambda)R}{\omega(1-\lambda)} \right) &\geq r^* \\ &\geq \max_{i \in \mathcal{I}'} \max \left(\frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)}, \right. \\ &\quad \left. \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right). \end{aligned} \quad (37)$$

Proof: Based on Formula (20), we have $\sum_i d_i = I'$, if

$$\begin{aligned} r^* &\geq \max_{i \in \mathcal{I}'} \max \left\{ \frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)}, \right. \\ &\quad \left. \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right\}. \end{aligned} \quad (38)$$

In the PT-based data forgetting game, the FL server aims to maximize its subjective expected utility. Thus, by (16), we have $\mathbb{U}_S^{PT}(1, \mathbf{r}^*) \geq \max \{ \mathbb{U}_S^{PT}(-1, \mathbf{r}^*), \mathbb{U}_S^{PT}(0, \mathbf{r}^*) \}$, if

$$\begin{cases} \theta_i G(1) \geq [\omega(1-\lambda) - \omega(\lambda)](R + r^*), \\ \theta_i G(1) \geq \omega(1-\lambda)r^* - \omega(\lambda)R. \end{cases} \quad (39)$$

yielding

$$r^* \leq \min \left(\frac{\theta_i G(1)}{\omega(1-\lambda) - \omega(\lambda)} - R, \frac{\theta_i G(1) + \omega(\lambda)R}{\omega(1-\lambda)} \right). \quad (40)$$

Therefore, by Formula (38) and Formula (40), we have

$$\begin{aligned} \min \left(\frac{\theta_i G(1)}{\omega(1-\lambda) - \omega(\lambda)}, \frac{\theta_i G(1) + \omega(\lambda)R}{\omega(1-\lambda)} \right) &\geq r^* \\ &\geq \max_{i \in \mathcal{I}'} \max \left(\frac{[\omega(\lambda) - \omega(1-\lambda)]R + C_i(1) - C_i(0)}{\omega(1-\lambda) - \omega(\lambda)}, \right. \\ &\quad \left. \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right). \end{aligned} \quad (41)$$

If Formula (41) holds, client i 's optimal strategy is $\mathbf{1}_{I'}$. Furthermore, as indicated by (16), the subjective expected utility of the FL server, i.e., $\mathbb{U}_S^{PT}$, is a strictly decreasing function with respect to r over the constraint in Formula (41). Consequently, the FL server's optimal reward within this constraint is the minimum possible value. Furthermore, considering the stability of the clients' responses, the FL server has no incentive to offer any reward r' greater than this minimum, as its subjective utility would decrease. Meanwhile, any reward $r' \neq r^*$ would cause the client's response to deviate from d^* . Therefore, $(\mathbf{d}^*, \mathbf{r}^*)$ is the optimal and unique game equilibrium. Theorem 1 is proved. ■

Remark: Theorem 1 indicates that when the FL server offers higher rewards, as shown in Formula (37), all clients are motivated to retain all data to participate in FUL. Furthermore, with zero-payment mechanism and penalty mechanism, the utility for a client providing a meaningless model is lower than that for a client who opts out of participating in FUL. Consequently, a client is improbable to contribute a meaningless model unless he/she is certain that a substantial number of other clients are simultaneously engaging in selfish behavior, which is unlikely to occur in a large-scale system.

Based on Theorem 1, we design a reward allocation algorithm grounded in PT-based data forgetting game for scenario I, as shown in Algorithm 1. Specifically, each client $i \in \mathcal{I}'$ first selects the optimal remaining data size and then engages in the FUL process in accordance with their chosen strategy. Subsequently, the FL server determines the optimal reward strategy for clients.

Computational complexity analysis: In Algorithm 1, lines 6-12 are to compute the optimal remaining data size for I' clients. The number of iterations required for this part is I' , leading to a computational complexity of $\mathcal{O}(I')$, where $\mathcal{O}$ is the big-O notation. Lines 14-18 iteratively determine the optimal reward for each remaining data size level to clients. The number of iterations required for this part is H , leading to a computational complexity of $\mathcal{O}(H)$. Accordingly, the overall computational complexity of Algorithm 1 is $\mathcal{O}(H + I')$.

B. Game Analysis in Scenario II ($H = 2$)

We further consider the equilibrium of the PT-based data forgetting game with $H = 2$ non-zero remaining data size levels. Thus, we have $\mathbf{r}^* = [0, -R, r_1^*, r_2^*]$. Proposition 2 proves a

Algorithm 1: Reward Allocation Algorithm for Scenario I.

```

1: Input:  $i \in \mathcal{I}'$ ,  $C_i(0)$ ,  $C_i(1)$ ,  $\lambda$ ,  $\alpha$ ,  $\theta$ ,  $G(1)$ ,  $R$ .
2: Output:  $d_i^*$ ,  $r_{d_i}^*$ .
3: The FL server designs a set of recruitment messages,
   which enumerate the reward associated with remaining
   data size, for clients previously involved in FL.
4: Stage I: Optimal remaining data size of the client
5: for ( $i = 1; i \leq I'; i++$ ) do
6:   if Formula (37) holds then
7:      $d_i^* = 1$ .
8:   else
9:      $d_i^* = -1$ .
10:  end if
11: end for
12: Stage II: Optimal reward strategy for the FL server
13: The FL server estimates the data size remaining in the
   system for each client.
14: for ( $i = 1; i \leq I'; i++$ ) do
15:   if  $\hat{d}_i = -1$  then
16:      $r_{d_i}^* = 0$ .
17:   else if  $\hat{d}_i = 0$  then
18:      $r_{d_i}^* = -R$ .
19:   else
20:      $r_{d_i}^* = \max_{i \in \mathcal{I}'} \max \left( \frac{-\Upsilon_2 R + C_i(1) - C_i(0)}{\Upsilon_2}, \frac{\omega(\lambda)R + C_i(1)}{\omega(1-\lambda)} \right)$ .
21:   end if
22: end for
23: Return  $d_i^*$ ,  $r_{d_i}^*$ .

```

quantitative correlation between the optimal reward strategy of the FL server and the strategy of the clients.

Proposition 2: In the PT-based data forgetting game with $H = 2$ positive data size levels, client i decides the maximum data size to remain in the system, i.e., $d_i = 2$, if

$$\begin{cases} r_2^* > \max \left\{ \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)}, \right. \\ \quad \left. \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\frac{\lambda}{2})R}{\omega(1-\lambda)^2 - \omega(\lambda)\omega(\frac{\lambda}{2})} \right\}, \\ \frac{\Upsilon_1 r_2^* + \omega(\frac{\lambda}{2})R + C_i(1) - C_i(2)}{\Upsilon_2} > r_1^* > \frac{C_i(2) - \omega(1-\lambda)r_2^*}{\omega(\lambda)}, \end{cases} \quad (42)$$

where $\Upsilon_1 = \omega(1-\lambda) - \omega(\frac{\lambda}{2})$ and $\Upsilon_2 = \omega(1-\lambda) - \omega(\lambda)$.

Client i decides the part of the data size to remain in the system, i.e., $d_i = 1$, if

$$\begin{cases} r_1^* > \max \left\{ \frac{\omega(1-\lambda)\omega(\frac{\lambda}{2})R + \omega(1-\lambda)C_i(1) - \omega(\frac{\lambda}{2})C_i(2)}{\omega^2(1-\lambda) - \omega(\lambda)\omega(\frac{\lambda}{2})}, \right. \\ \quad \left. \frac{[2\Pi - \omega^2(1-\lambda)]R + \omega(1-\lambda)C_i(1) - \omega(\frac{\lambda}{2})C_i(2) - \Upsilon_1 C_i(0)}{\omega(1-\lambda)\Upsilon_1} \right\}, \\ \frac{\Upsilon_2 r_1^* + C_i(2) - C_i(1) - \omega(\frac{\lambda}{2})R}{\Upsilon_1} > r_2^* > \\ \quad \max \left\{ R + \frac{C_i(1) - \omega(1-\lambda)r_1^*}{\omega(\frac{\lambda}{2})}, \frac{C_i(1) - C_i(0) - \Upsilon_2 r_1^* - \Upsilon_1 R}{\omega(\frac{\lambda}{2})} \right\}, \end{cases} \quad (43)$$

where $\Pi = \omega(1-\lambda)\omega(\frac{\lambda}{2})$.

Client i is stimulated to revoke all data through FUL, i.e., $d_i = -1$, if

$$\begin{cases} r_1^* < \frac{\omega(1-\lambda)R + C_i(0)}{\omega(\lambda)}, \\ r_2^* < \min \left\{ \frac{C_i(1) + \omega(\frac{\lambda}{2})R - \omega(1-\lambda)r_1^*}{\omega(\lambda)}, \frac{C_i(2) - \omega(\lambda)r_1^*}{\omega(1-\lambda)} \right\}. \end{cases} \quad (44)$$

Proof: As $H = 2$, (6) can be reformulated as

$$\Pr(\hat{d}|d) = \begin{cases} 1-\lambda, & \text{if } \hat{d} = d \geq 0, \\ \lambda, & \text{if } (d, \hat{d}) = (0, 1) \text{ or } (2, 1), \\ \lambda/2, & \text{if } \hat{d} = d \pm 1, 0 < q < H, \\ 1, & \text{if } (d, \hat{d}) = (-1, -1), \\ 0, & \text{otherwise.} \end{cases} \quad (45)$$

Based on (17) and (45), we have

$$\begin{aligned} \mathbb{U}_i^{PT}(2, r^*) &= \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^* - C_i(2) \\ &\geq 0 = \mathbb{U}_i^{PT}(-1, r^*), \end{aligned} \quad (46)$$

if

$$r_1^* \geq \frac{C_i(2) - \omega(1-\lambda)r_2^*}{\omega(\lambda)}, \forall i \in \mathcal{I}', \quad (47)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(2, r^*) &= \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^* - C_i(2) \\ &\geq -\omega(1-\lambda)R + \omega(\lambda)r_1^* - C_i(0) = \mathbb{U}_i^{PT}(0, r^*), \end{aligned} \quad (48)$$

if

$$r_2^* \geq \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)}, \forall i \in \mathcal{I}', \quad (49)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(2, r^*) &= \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^* - C_i(2) \\ &\geq \omega(1-\lambda)r_1^* + \omega\left(\frac{\lambda}{2}\right)r_2^* - \omega\left(\frac{\lambda}{2}\right)R - C_i(1) = \mathbb{U}_i^{PT}(1, r^*), \end{aligned} \quad \text{if} \quad (50)$$

if

$$r_1^* \leq \frac{\Upsilon_1}{\Upsilon_2}r_2^* + \frac{\omega\left(\frac{\lambda}{2}\right)R + C_i(1) - C_i(2)}{\Upsilon_2}, \forall i \in \mathcal{I}'. \quad (51)$$

Therefore, combining Formula (47), Formula (49), and Formula (51), we have

$$\mathbb{U}_i^{PT}(2, r^*) \geq \max \{ \mathbb{U}_i^{PT}(-1, r^*), \mathbb{U}_i^{PT}(0, r^*), \mathbb{U}_i^{PT}(1, r^*) \}, \quad (52)$$

if

$$\begin{cases} r_2^* > \max \left\{ \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)}, \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega(1-\lambda)^2 - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)} \right\}, \\ \frac{\Upsilon_1 r_2^* + \omega\left(\frac{\lambda}{2}\right)R + C_i(1) - C_i(2)}{\Upsilon_2} > r_1^* > \frac{C_i(2) - \omega(1-\lambda)r_2^*}{\omega(\lambda)}. \end{cases} \quad (53)$$

Therefore, if Formula (42) holds, client i is stimulated to decide the maximum data size to remain in the system, i.e., $d_i = 2$.

Similarly, from (17) and (45), we have

$$\begin{aligned} \mathbb{U}_i^{PT}(1, r^*) &= \omega(1-\lambda)r_1^* + \omega\left(\frac{\lambda}{2}\right)r_2^* - \omega\left(\frac{\lambda}{2}\right)R - C_i(1) \\ &\geq 0 = \mathbb{U}_i^{PT}(-1, r^*), \end{aligned} \quad (54)$$

if

$$r_2^* \geq \frac{\omega\left(\frac{\lambda}{2}\right)R + C_i(1) - \omega(1-\lambda)r_1^*}{\omega\left(\frac{\lambda}{2}\right)}, \forall i \in \mathcal{I}', \quad (55)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(1, r^*) &= \omega(1-\lambda)r_1^* + \omega\left(\frac{\lambda}{2}\right)r_2^* - \omega\left(\frac{\lambda}{2}\right)R - C_i(1) \\ &\geq -\omega(1-\lambda)R + \omega(\lambda)r_1^* - C_i(0) = \mathbb{U}_i^{PT}(0, r^*), \end{aligned} \quad (56)$$

if

$$r_2^* \geq \frac{C_i(1) - \Upsilon_2 r_1^* - C_i(0) - \Upsilon_1 R}{\omega\left(\frac{\lambda}{2}\right)}, \forall i \in \mathcal{I}', \quad (57)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(1, r^*) &= \omega(1-\lambda)r_1^* + \omega\left(\frac{\lambda}{2}\right)r_2^* - \omega\left(\frac{\lambda}{2}\right)R - C_i(1) \\ &\geq \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^* - C_i(2) = \mathbb{U}_i^{PT}(2, r^*), \end{aligned} \quad (58)$$

if

$$r_2^* \leq \frac{\Upsilon_2}{\Upsilon_1}r_1^* + \frac{C_i(2) - C_i(1) - \omega\left(\frac{\lambda}{2}\right)R}{\Upsilon_1}, \forall i \in \mathcal{I}'. \quad (59)$$

Therefore, combining Formula (55), Formula (57), and Formula (59), we have

$$\mathbb{U}_i^{PT}(1, r^*) \geq \max \{ \mathbb{U}_i^{PT}(-1, r^*), \mathbb{U}_i^{PT}(0, r^*), \mathbb{U}_i^{PT}(2, r^*) \}, \quad (60)$$

$$\begin{cases} r_1^* > \max \left\{ \frac{\omega(1-\lambda)\omega\left(\frac{\lambda}{2}\right)R + \omega(1-\lambda)C_i(1) - \omega\left(\frac{\lambda}{2}\right)C_i(2)}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)}, \frac{[2\Pi - \omega^2(1-\lambda)]R + \omega(1-\lambda)C_i(1) - \omega\left(\frac{\lambda}{2}\right)C_i(2) - \Upsilon_1 C_i(0)}{\omega(1-\lambda)\Upsilon_1} \right\}, \\ \frac{\Upsilon_2 r_1^* + C_i(2) - C_i(1) - \omega\left(\frac{\lambda}{2}\right)R}{\Upsilon_1} > r_2^* > \max \left\{ R + \frac{C_i(1) - \omega(1-\lambda)r_1^*}{\omega\left(\frac{\lambda}{2}\right)}, \frac{C_i(1) - C_i(0) - \Upsilon_2 r_1^* - \Upsilon_1 R}{\omega\left(\frac{\lambda}{2}\right)} \right\}. \end{cases} \quad (61)$$

Therefore, if Formula (43) holds, client i is stimulated to decide the part of the data size to remain in the system, i.e., $d_i = 1$.

Similarly, from (17) and (45), we have

$$\begin{aligned} \mathbb{U}_i^{PT}(-1, r^*) &= 0 \geq -\omega(1-\lambda)R + \omega(\lambda)r_1^* - C_i(0) \\ &= \mathbb{U}_i^{PT}(0, r^*), \end{aligned} \quad (62)$$

if

$$r_1^* \leq \frac{C_i(0) + \omega(1-\lambda)R}{\omega(\lambda)}, \forall i \in \mathcal{I}', \quad (63)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(-1, r^*) &= 0 \\ &\geq \omega(1-\lambda)r_1^* + \omega\left(\frac{\lambda}{2}\right)r_2^* - \omega\left(\frac{\lambda}{2}\right)R - C_i(1) = \mathbb{U}_i^{PT}(1, r^*), \end{aligned} \quad (64)$$

if

$$r_2^* < \frac{C_i(1) + \omega\left(\frac{\lambda}{2}\right)R - \omega(1-\lambda)r_1^*}{\omega\left(\frac{\lambda}{2}\right)}, \forall i \in \mathcal{I}', \quad (65)$$

and

$$\begin{aligned} \mathbb{U}_i^{PT}(-1, r^*) &= 0 \\ &\geq \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^* - C_i(2) = \mathbb{U}_i^{PT}(2, r^*), \end{aligned} \quad (66)$$

if

$$r_2^* < \frac{C_i(2) - \omega(\lambda)r_1^*}{\omega(1-\lambda)}, \forall i \in \mathcal{I}'. \quad (67)$$

Hence, if Formula (44) holds, client i is stimulated to revoke all data, i.e., $d_i = -1$. Proposition 2 is proved. ■

Remark: As demonstrated in Formula (42), to enhance the performance of the unlearned model, the FL server is required to allocate a higher reward to the client that decides a greater remaining data size r_2^* than that to the client that decides a lesser remaining data size r_1^* . If condition Formula (43) holds, i.e., r_1^* is excessively high, client i is motivated to choose a smaller remaining data size. Furthermore, if the cost associated with executing the FUL task surpasses the rewards r_1^* and r_2^* given by Formula (44), clients will be inclined to revoke all data.

The equilibrium of the PT-based data forgetting game with $H = 2$ non-zero remaining data size levels is proved in the following Theorem 2.

Theorem 2: In the PT-based data forgetting game with $H = 2$ positive remaining data size levels and I' clients, the game equilibrium is given by

$$\mathbf{d}^* = \mathbf{2}_{I'}, \quad (68)$$

$$\begin{aligned} \mathbf{r}^* &= \left[0, -R, \max_{i \in \mathcal{I}'} \frac{C_i(2) - \omega(1-\lambda)r_2^*}{\omega(\lambda)} \right], \\ &\max_{i \in \mathcal{I}'} \max \left\{ \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)}, \right. \\ &\quad \left. \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)} \right\} \Big], \end{aligned} \quad (69)$$

if

$$\begin{aligned} &\frac{\theta_i[\omega(1-\lambda)G(2) - \omega(\lambda)G(1)] - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)} \geq r_2^* \\ &\geq \max \left\{ \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)}, \right. \\ &\quad \left. \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)} \right\}. \end{aligned} \quad (70)$$

Proof: Similar to Theorem 1, according to Formula (42), we have $\sum_i d_i = 2I'$, if

$$\begin{cases} r_2^* > \max_{i \in \mathcal{I}'} \max \left\{ \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)}, \right. \\ \quad \left. \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega(1-\lambda)^2 - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)} \right\}, \\ \min_{i \in \mathcal{I}'} \frac{\Upsilon_1 r_2^* + \omega\left(\frac{\lambda}{2}\right)R + C_i(1) - C_i(2)}{\Upsilon_2} > r_1^* > \max_{i \in \mathcal{I}'} \frac{C_i(2) - \omega(1-\lambda)r_2^*}{\omega(\lambda)}. \end{cases} \quad (71)$$

In the PT-based data forgetting game, the FL server aims to maximize its subjective expected utility. Thus, by (16), we have $\mathbb{U}_S^{PT}(2, \mathbf{r}^*) \geq \max\{\mathbb{U}_S^{PT}(-1, \mathbf{r}^*), \mathbb{U}_S^{PT}(0, \mathbf{r}^*), \mathbb{U}_S^{PT}(1, \mathbf{r}^*)\}$, if

$$\begin{cases} \theta_i G(2) \geq \omega(\lambda)r_1^* + \omega(1-\lambda)r_2^*, \\ \theta_i G(2) \geq \omega(1-\lambda)r_2^* + \omega(1-\lambda)R, \\ \theta_i [G(2) - G(1)] \geq \omega\left(\frac{\lambda}{2}\right)R - \Upsilon_2 r_1^* + \Upsilon_1 r_2^*. \end{cases} \quad (72)$$

yielding

$$\frac{\theta_i[\omega(1-\lambda)G(2) - \omega(\lambda)G(1)] - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)} \geq r_2^*. \quad (73)$$

Based on Formula (71) and Formula (73), we can obtain

$$\begin{aligned} &\frac{\theta_i[\omega(1-\lambda)G(2) - \omega(\lambda)G(1)] - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)} \geq r_2^* \\ &\geq \max \left\{ \frac{\omega(1-\lambda)C_i(2) - \omega(\lambda)C_i(1) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)R}{\omega^2(1-\lambda) - \omega(\lambda)\omega\left(\frac{\lambda}{2}\right)}, \right. \\ &\quad \left. \frac{C_i(2) - C_i(0) - \omega(1-\lambda)R}{\omega(1-\lambda)} \right\}. \end{aligned} \quad (74)$$

Hence, if Formula (70) holds, the optimal strategy of client i is $\mathbf{2}_{I'}$. Furthermore, as indicated by (16), the subjective expected utility of the FL server, i.e., $\mathbb{U}_S^{PT}$, is a strictly decreasing function with respect to r . Consequently, the optimal strategy of the FL server is the smallest positive solution that fulfills the condition in Formula (70). Therefore, we can obtain the specific form of the r^* in (69). Theorem 2 is proved. ■

Remark: As shown in Theorem 2, when condition Formula (70) holds, the FL server compensates all clients that choose a larger remaining data size by providing higher rewards as indicated in (69). Moreover, Theorem 2 shows that the FL server's optimal reward strategy decreases as the degree of rationality increases in order to pursue higher profits.

Based on Theorem 2, we propose a reward allocation algorithm grounded in the PT-based data forgetting game for scenario II, as shown in Algorithm 2, to obtain the equilibrium for the data forgetting game with $H = 2$ positive remaining data size levels. The complexity of Algorithm 2 is the same as that of Algorithm 1, which is $\mathcal{O}(H + I')$.

The equilibrium for the general PT-based data forgetting game with H positive data size levels and I' clients is denoted as $(\mathbf{H}_{I'}, \mathbf{r}^*)$. According to definition, for $i \in \mathcal{I}'$, we have

$$\mathbb{U}_S^{PT}(\mathbf{H}_{I'}, \mathbf{r}^*) \geq \mathbb{U}_S^{PT}([d_i]_{i \in \mathcal{I}'}, \mathbf{r}^*), \forall -1 \leq d_i \leq H. \quad (75)$$

$$\mathbb{U}_i^{PT}(H, \mathbf{r}^*) \geq \max_{-1 \leq h \leq H-1} \mathbb{U}_i^{PT}(h, \mathbf{r}^*). \quad (76)$$

Algorithm 2: Reward Allocation Algorithm for Scenario II.

```

1: Input:  $i \in \mathcal{I}'$ ,  $C_i(0)$ ,  $C_i(1)$ ,  $C_i(2)$ ,  $\lambda$ ,  $\alpha$ ,  $\theta_i$ ,  $G(1)$ ,  $G(2)$ ,  $R$ .
2: Output:  $d_i^*$ ,  $r_{d_i}^*$ .
3: The FL server designs a set of recruitment messages,
   which enumerate the reward associated with remaining
   data size, for clients previously involved in FL.
4: Stage I: Optimal remaining data size of the client
5: for ( $i = 1$ ;  $i \leq I'$ ;  $i++$ ) do
6:   if Formula (70) holds then
7:      $d_i^* = 2$ .
8:   else
9:      $d_i^* = -1$ .
10:  end if
11: end for
12: Stage II: Optimal reward strategy for the FL server
13: The FL server estimates the data size remaining in the
   system for each client.
14: for ( $i = 1$ ;  $i \leq I'$ ;  $i++$ ) do
15:   if  $\hat{d}_i = -1$  then
16:      $r_{d_i}^* = 0$ .
17:   else if  $\hat{d}_i = 0$  then
18:      $r_{d_i}^* = -R$ .
19:   else
20:      $r_{d_i}^* = \hat{h}$ , where  $\hat{h}$  is the fourth element in (69).
21:   end if
22: end for
23: Return  $d_i^*$ ,  $r_{d_i}^*$ .

```

When the profit derived from a high remaining data size is substantial, the FL server is motivated to offer a high reward, thereby encouraging each staying client to choose the maximum remaining data size to engage in the FUL process. Conversely, in the absence of such incentives, clients may choose not to perform the FUL tasks or even choose to contribute meaningless models for profit.

Computational overhead analysis: Within each global unlearning round, the overall computational overhead of the proposed scheme is $\mathcal{O}(H + 2I' + \max\{Z_i, i \in \mathcal{I}'\})$. Specifically, this overhead is broken down into the following five aspects: 1) The FL server executes Algorithm 1 or Algorithm 2 to design a set of remaining data size-reward pairs for the staying clients, which are then broadcast to all staying clients. This aspect incurs a computational complexity of $\mathcal{O}(H + I')$. 2) The FL server removes the historical updates of the clients that ultimately leave the system and retrieves the most recent global model. This aspect incurs a computational complexity of $\mathcal{O}(1)$. 3) Each staying client i trains its local model using the gradient descent algorithm based on its chosen data size, where the number of local training epochs is Z_i . The computational complexity of this aspect is $\mathcal{O}(\max\{Z_i, i \in \mathcal{I}'\})$. 4) The FL server evaluates the remaining data size of each staying client via the remaining data size assessment model. The computational complexity of this aspect is $\mathcal{O}(I')$. 5) The FL server aggregates the local models provided by the staying clients, resulting in a computational complexity of $\mathcal{O}(1)$.

The remaining data assessment requires each staying client to execute one additional local unlearning round before the formal FUL process. The additional client-side computation incurred by the proposed assessment is $\mathcal{O}(\max\{Z_i, i \in \mathcal{I}'\})$.

Communication overhead analysis: Within each global unlearning round, the overall communication overhead of the proposed scheme is $(\vartheta + I'\zeta)$ MB + $(\mathbb{E} + \mathbb{K})$ bits. Specifically, this overhead is broken down into the following four aspects: 1) The FL server broadcasts a set of optimal recruitment strategies to all staying clients. Assuming the strategy set size is $\mathbb{E}$ bits, the communication overhead for this part is $\mathbb{E}$ bits. 2) The FL server broadcasts the global model to all staying clients. Denoting the model size as ϑ MB, this results in a communication overhead of ϑ MB. 3) Each staying client uploads its trained local model to the FL server. Similar to [33], we assume all local models have the same size ζ MB, resulting in a communication overhead of $I'\zeta$ MB. 4) The FL server distributes rewards to the staying clients based on the evaluated remaining data sizes. Denoting the payment file size as $\mathbb{K}$ bits, this results in a communication overhead of $\mathbb{K}$ bits.

The proposed scheme only requires one additional model transmission per staying client, i.e., the upload of the parameters obtained after the extra local unlearning round. Hence, the additional communication cost is $I'\zeta$ MB. For a concrete quantitative measurement, taking ResNet-18 (≈ 11.7 M parameters) as an example, the extra transmission corresponds to approximately 46 MB per client.

In terms of communication overhead, relative to conventional FUL methods, the proposed scheme incorporates supplementary operations involving the dissemination of optimal recruitment strategies and the distribution of rewards. Given that the sizes of both recruitment strategies and the reward files are substantially smaller than those of the local and global models, the added communication burden from these elements is marginal. Regarding computational complexity, the computational cost of the proposed scheme increases linearly with the client population. Thus, the proposed scheme remains highly efficient even under large-scale client participation.

VII. PERFORMANCE EVALUATION

In this section, the experimental setup is first introduced, followed by the simulation results and discussions.

A. Experimental Setup

Parameters: We consider a classic FUL architecture, consisting of a FL server and $I = 10$ clients. These clients have participated in FL and are now simultaneously determining the amount of data to be revoked from the learned model to collaboratively perform the unlearning task. In the PT-based data forgetting game with $H = 1$, client i 's training cost with a remaining data size $d_i = 1$ is set as $C_i(1) \sim \Gamma(8, 12)$, where $\Gamma(x, y)$ denotes the uniform distribution over the interval $[x, y]$. Client i 's training cost with a remaining data size $d_i = 0$ is set as $C_i(0) \sim \Gamma(1, 2)$ [23]. Following the commonly adopted practice in PT-based mechanism design [34], [35], [36], we assign the behavioral parameters within a reasonable range, i.e., $\alpha \in [0.7, 0.9]$, and assess performance under varying settings

within this interval⁸. The evaluation error of the FL server is randomly selected within $[0.1, 0.3]$. The weighting parameter for the FL server's satisfaction is set to 10.

Datasets and Model: We use three commonly used image classification datasets to evaluate the performance of the truthful incentive scheme, including MNIST [37], CIFAR-10 [38], and Tiny-ImageNet [39]. To better reflect the effects of data distribution in practical FUL applications, we assess the performance of the proposed scheme on both IID and non-IID scenarios. For model training, a customized deep convolutional neural network (CNN) is adopted for MNIST and CIFAR-10, whereas the ResNet-18 architecture is implemented to handle the higher dimensionality of the Tiny-ImageNet dataset. In addition, the learning rate is set to 0.001, with a local batch size of 64. The total number of global unlearning rounds is set to 50.

In addition, to avoid systematically changing a client's local data distribution during the FUL, each client performs data deletion by uniform random sampling over its local dataset. In the proposed scheme, we consider three data-unlearning strategies for each client in the FUL process:

- 1) *Full Unlearning*: The client completely withdraws its local dataset and no longer participates in the FUL.
- 2) *Partial Unlearning*: The client uniformly and randomly samples 50% of its local data⁹, and only employs this retained subset for subsequent model training and uploading. A consistent uniform sampling procedure is used throughout to ensure that the client's original data distribution does not shift in expectation.
- 3) *Non-Unlearning*: The client does not remove any samples and continues to participate in FUL using its full local dataset.

We evaluate the effectiveness of the proposed truthful incentive scheme by comparing it with the following benchmarks.

- *Auction scheme (AS)* [40]: In AS, the FL server determines its optimal reward strategy based on the auction game. Each client chooses its optimal remaining data size strategy based on the auction game.
- *Random scheme (RS)*: In RS, the FL server randomly offers rewards to motivate clients to execute the FUL tasks.

B. Simulation Results

Fig. 2 shows the evolution of the optimal reward with different levels of rationality. Besides, three different values of evaluation error $\lambda = \{0.20, 0.25, 0.30\}$ is utilised to show results. As seen in Fig. 2, we can observe that the optimal reward for the FL server decreases as the level of rationality increases. The reason is that a higher level of rationality enables the FL server to

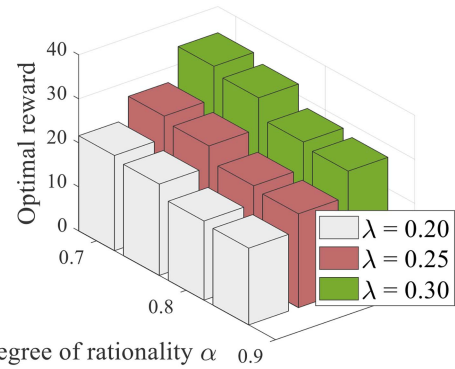


Fig. 2. Evolution of the optimal reward with different levels of rationality.

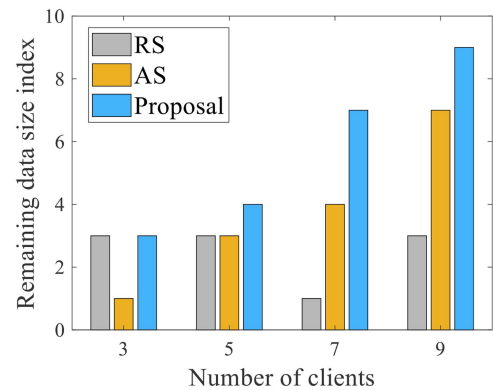


Fig. 3. Comparative analysis of the proposed scheme and benchmarks in terms of the remaining data size index.

make more objective decisions, thereby reducing the optimal reward paid to clients to achieve higher returns. Furthermore, for a fixed level of rationality, the optimal reward for the FL server increases with higher evaluation error. This is due to the fact that as the evaluation error of the FL server increases, it becomes more challenging for the FL server to accurately evaluate the clients' selected remaining data sizes, resulting in an inability to provide appropriate rewards to clients. Consequently, greater compensation is required to account for the FL server's misestimation of clients' remaining data sizes, leading to an increase in the optimal reward, which aligns with the theoretical analysis in Proposition 1.

Fig. 3 presents a comparative analysis of the proposed scheme and benchmarks in terms of the remaining data size index for the FUL system. In our paper, the remaining data size index for the FUL system refers to the cumulative sum of the remaining data sizes received by the FL server from the staying clients. From Fig. 3, we can observe that the proposed scheme outperforms benchmarks in terms of this index. The reasons are as follows. In AS, the FL server selects the lowest-bidding clients to perform the unlearning task, without considering the remaining data size chosen by the client. This leads to the possibility that clients may receive higher rewards even if they opt to revoke a larger amount of data. Thus, the remaining data size index is not globally optimal. In RS, the FL server randomly allocates rewards to

⁸In evolving FUL systems, the FL server can dynamically estimate the behavioral parameter by analyzing historical interaction records with clients. Specifically, the FL server can compare the estimated remaining data size with the corresponding reference values (e.g., obtained through accumulated interaction records) and adjust α based on the observed estimation error.

⁹Note that our scheme is based on uniform random sampling, which inherently supports arbitrary retention granularities (e.g., 20%, 80%). By using uniform sampling, we ensure that the expected data distribution remains unchanged regardless of the retention ratio. Therefore, while we report the 50% case for conciseness, the proposed approach generalizes effectively to various levels of data granularity without introducing distributional bias.

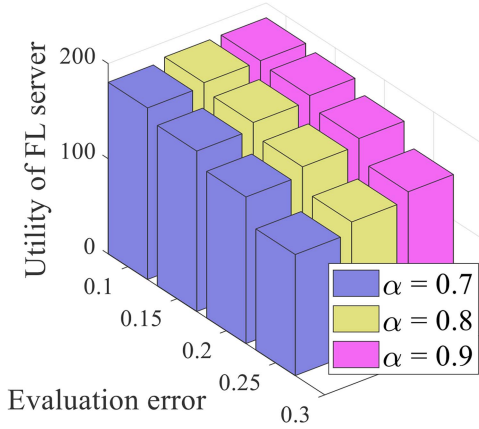


Fig. 4. Evolution of the utility of the FL server with different evaluation errors.

clients, which may lead to clients either not participating in the FUL process or providing meaningless models, resulting in a lower remaining data size index. In the proposed scheme, guided by the PT-based data forgetting game, the FL server offers optimal rewards based on the remaining data size provided by each client, thereby incentivizing clients to retain more data for model training. Additionally, with the zero-payment and penalty mechanism, the selfish behaviors of clients can be effectively mitigated. Consequently, the remaining data size index is maximized.

Fig. 4 shows the impact of evaluation error on the utility of the FL server, where three different levels of rationality value $\alpha = \{0.70, 0.85, 0.90\}$ are exploited. In Fig. 4, the utility of the FL server decreases as the evaluation error increases. This is because a higher evaluation error forces the FL server to allocate greater rewards to compensate clients, thereby motivating their participation in the FUL process. Consequently, the FL server's utility is reduced due to the increased compensation costs. Moreover, for a fixed evaluation error, the utility of the FL server increases with the increase of rationality. The reason is that a higher rationality enables the FL server to minimize the risk of overpayment, thereby increasing the FL server's utility. The observed phenomenon is consistent with the theoretical analysis presented in Theorem 2.

Fig. 5 shows a comparative analysis of the proposed scheme and benchmarks in terms of the average utility for clients. In Fig. 5, the proposed scheme outperforms the benchmarks in attaining a higher average utility. This can be interpreted as follows. In AS, the FL server allocates rewards based solely on the lowest bids of clients, without considering client heterogeneity, which results in a lower average utility for clients. In RS, the strategies of the FL server and clients are random, leading to a lower average utility for clients. In the proposed scheme, the FL server leverages the PT-based data forgetting game to design optimal recruitment strategies, ensuring that clients receive optimal rewards commensurate with their contributions. Therefore, the average utility for clients is maximized.

Figs. 6 and 7 present a comparative analysis of the training loss across the three schemes for the MNIST and CIFAR-10 datasets. From Figs. 6 and 7, the proposed scheme outperforms the

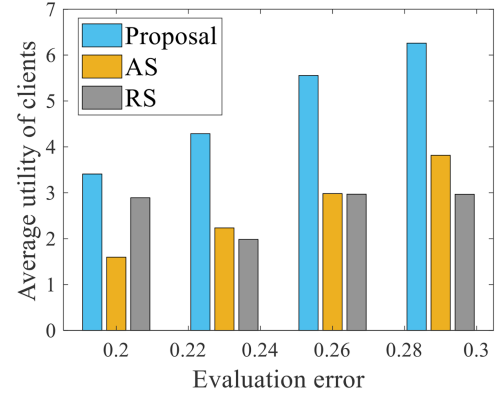


Fig. 5. Comparative analysis of the proposed scheme and benchmarks in terms of the average utility for clients.

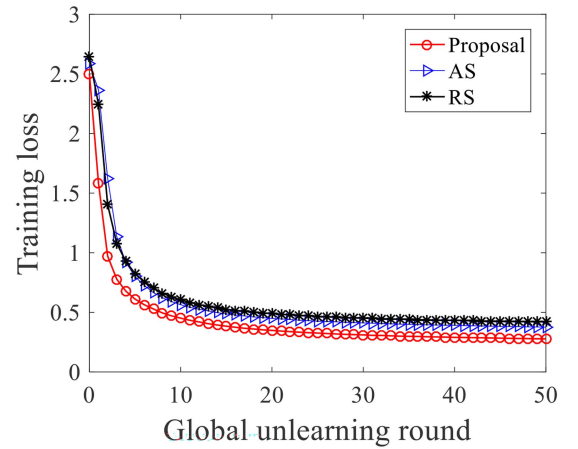


Fig. 6. Comparison of the proposed scheme and benchmarks in terms of the training loss under non-IID conditions (MNIST).

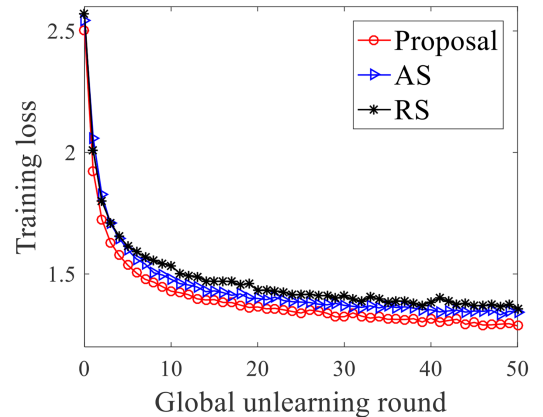


Fig. 7. Comparison of the proposed scheme and benchmarks in terms of the training loss under non-IID conditions (CIFAR-10).

benchmarks in terms of the training loss on both the MNIST and CIFAR-10 datasets. This observation underscores the efficacy of our proposed scheme in motivating clients to retain more data during the unlearning process, even in scenarios where the data exhibits non-IID characteristics. The reason is that

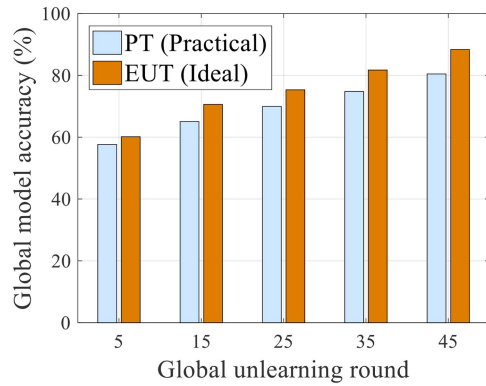


Fig. 8. Comparative analysis of the proposed scheme under PT and EUT in terms of the global model accuracy (MNIST).

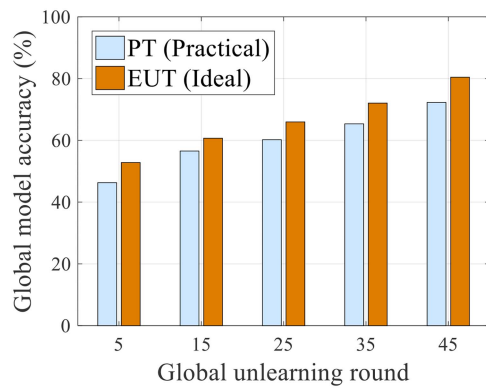


Fig. 9. Comparative analysis of the proposed scheme under PT and EUT in terms of the global model accuracy (CIFAR-10).

the proposed scheme effectively mitigates the selfish behavior of clients through the implementation of the zero-payment and penalty mechanisms. Furthermore, the FL server provides optimal rewards for clients by leveraging the PT-based data forgetting game, thereby effectively encouraging them to contribute high-quality models during the FUL process, resulting in minimal training loss.

To compare the performance of the proposed scheme under practical conditions (i.e., PT) versus ideal conditions (i.e., EUT), we have added experiments that evaluate the global model accuracy under both PT and EUT. Figs. 8 and 9 present a comparative analysis of the proposed scheme under PT and EUT in terms of the global model accuracy on the MNIST and CIFAR-10 datasets. As shown in Figs. 8 and 9, the EUT-based scheme achieves higher global model accuracy compared with the PT-based scheme, which can be explained as follows. In the EUT-based scheme (i.e., the ideal case), the FL server can objectively evaluate the amount of data retained by each client and determine the optimal payment strategy based on this evaluation under the guidance of EUT. Consequently, clients are incentivized to retain more data, leading to the maximum global model accuracy. In the PT-based scheme (i.e., the practical case), the FL server with bounded rationality cannot objectively and accurately assess the clients' retained data size, which may result in situations where clients receive higher payments even when

TABLE II
IMPACT OF THE PROPOSED PT-BASED INCENTIVE SCHEME ACROSS THREE UNLEARNING SOLUTIONS ON CIFAR-10 AND TINY-IMAGENET (ACCURACY)

Dataset	Method	IFU	PRM	HFU
CIFAR-10	W/ incentive	81.67%	79.92%	74.64%
	W/O incentive	78.44%	74.17%	70.55%
Tiny-ImageNet	W/ incentive	45.17%	39.97%	36.32%
	W/O incentive	41.58%	36.13%	31.06%

they forget a larger portion of their data. As a result, the global model accuracy is not globally optimal. This reflects the negative impact of the FL server's bounded rationality on the performance of the FUL system.

To demonstrate the generalizability of the proposed incentive scheme, we evaluate the performance of our PT-based incentive scheme when integrated with different unlearning solutions, including Hessian-based FUL (HFU) [41], influence-based FUL (IFU) [42], and partial retraining methods (PRM) [43]. As shown in Table II, the proposed incentive scheme remains effective across these different unlearning methods. This is because [41], [42], [43] focus on designing optimized unlearning methods to improve the efficiency of the unlearning process, without considering the economic and strategic aspects of FUL. Through our incentive scheme, regardless of the chosen unlearning method, the FL server pays corresponding rewards based on the size of the remaining data utilized by clients, thereby motivating clients to actively participate in unlearning and provide high-quality services. In addition, Table II demonstrates that the proposed scheme maintains high model accuracy even as the number of classes and model complexity increase.

VIII. CONCLUSION

This paper has proposed a strategic data forgetting scheme with bounded rationality designed to encourage client participation in FUL while mitigating selfish behaviors through the design of optimal incentives. Specifically, a practical FUL framework has been first developed, featuring a bounded rational FL server that allows clients to revoke part of their data. Additionally, a PT-based data forgetting game has been formulated to analyze the optimal incentive for the FL server and the corresponding data revocation strategies for clients, with PT applied to model the FL server's decision-making process under conditions of risk and uncertainty. Furthermore, we have proposed a penalty mechanism to suppress clients' selfish behavior and a zero-payment mechanism to ensure fair reward allocation. Finally, simulation results have validated the efficacy of the proposed scheme in motivating client participation in FUL, curbing selfish behavior, and enhancing the unlearned model's accuracy. For future work, to enhance resilience, we will further extend our research to robust FUL incentive schemes to enhance system security against colluding clients.

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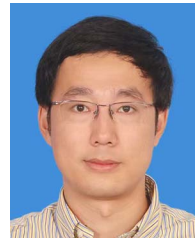
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